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Gamifying Education: Motivation and the Implementation of Digital Badges for use in Higher Education

As early as the Revolutionary War, soldiers were provided with emblems or "badges" for their bravery and good conduct (McAfee, 2015). Even today, the military continues to wear crests on their uniforms symbolizing vigilance and valor. Badges are also used to certify knowledge and learning (Halavais, 2012). In fact, the first Boy Scout badges, made of cloth with an embroidered design, were produced in 1911 to signify evidence of proficiency or merit in a particular area like agriculture (Wills, 2009). Today, there are over a hundred badges in various areas and skill levels. These visual representations of credentials are symbolic and highlight the qualifications and characteristics of those wearing them.

During the first half of the twentieth century, the use of rewards became evident in the field of education. Based upon operant learning theory, the use of rewards such as candy, stickers, badges, and grades have been used to extrinsically motivate students. The premise was to promote a desired behavior. And, according to behaviorist B. F. Skinner (1950), the use of rewards strengthens a student's behavior, especially if the individual finds merit in the incentive system.

Scholars have suggested that rewards may also motivate individuals to pursue more challenging tasks. For example, online gaming is growing at a rapid pace. Participants earn rewards such as badges, points, and achievements for accomplishing a particular level in the game. Begy and Consalvo (2011) stated that "games are designed to continually reinforce the player's position in the fictional world" (para. 2). Tom Chatfield (2010) noted that "games are brilliant at this... every time you do something, you get credit; you get a credit for trying" (para. 8). Also, Kapp (2012), noted that these rewards provide social capital and bragging rights.

Additionally, Suh, Wagner, and Liu (2016) reported that gamification enhanced user psychological needs satisfaction.

Recently, components from gaming are being implemented into education in the form of digital badges (Foster, 2013). Originally static graphical images, today's digital badges are "web-enabled tokens of accomplishment that contain specific claims and evidence about learning and achievement along with detailed evidence supporting those claims" (O'Byrne, Schenke, Willis, & Hickey, 2015, p. 451). Furthermore, the practice of creating, rewarding, and sharing digital badges has emerged as a means to motivate and reward student learning. Digital badges may identify "anytime, anywhere learning" as students acquire credentials that actually measure skills, competencies, and achievements obtained in the classroom, workforce, and community (USDE, 2011).

Related Literature

The MacArthur Foundation (2017) defined a digital badge as "an assessment and credentialing mechanism that is housed and managed online. Badges are designed to make visible and validate learning in both formal and informal settings, and hold the potential to help transform where and how learning is valued" (para. 1). Simply stated, a digital badge is a reward but also a visual symbol of ones' credentials (Otto & Hickey, 2014). According to Delello and McWhorter (2015), a digital badge is comprised of micro-credentials including the purpose of the badge, the date the badge was awarded, who issued the badge, and who earned the badge. Digital badges are verifiable, stackable, and portable for sharing across social media sites (Mozilla, 2016). In addition, the ability to display digital badges may "induce competition among badge earners" (Schenke, Tran, & Hickey, 2013, para. 8).

According to Twarnoite (2015), 75% of the Millennial Generation (defined as adults 18-34) will be part of the global workforce by 2025. A recent study by the University Professional and Continuing Education Association (UPCEA, 2017) indicated that millennials valued alternative credentialing in the form of certificates and digital badges. Similarly, Zalaznick (2016) noted that in higher education, it is the millennial generation driving the badging system in order to earn microcredentials for the workforce.

The Use of Badges across Education

The use of digital badges remained largely untapped until 2011 when Mozilla and The MacArthur Foundation joined forces to explore the digital badge movement. The organizations sponsored a two million dollar competition for the creation and assessment of digital badges (MacArthur Foundation, 2011). Various leaders across the United States participated in the initiative. For example, at the launch of the competition, Secretary Arne Duncan of the U.S. Department of Education stated, “Badges can help engage students in learning, and broaden the avenues for learners of all ages to acquire and demonstrate – as well as document and display – their skills” (USDE, 2011, para. 12)..

In 2012, the Harvard Review reported that digital badges were one of four innovative trends to watch in 2013 (Schrage, 2012). The report suggested that the system we currently use of accreditation may favor badges over traditional grades and digital diplomas to show skills acquisition and course completion. Even the non-profit Kahn Academy, which is well-known for its free content based instructional micro-lectures, integrated a system of points and badges into its learning resource system. From meteorite (see Figure 1) to moon badges, students earn badges for completing challenges, obtaining, points, watching videos, answering questions, or even donating to Kahn (Kahn Academy, 2015).

Figure 1. Kahn Academy Earth Badge. (© 2017, Kahn Academy. Used with permission.).

One of the leading learning management systems (LMS), Blackboard has also taken a key role in the production and disbursement of digital badges. A recent Blackboard blog stated, "Digital badging has risen as a "common currency" that transcends multiple learning venues… such as MOOCs, professional development courses and workplace training and allows students to display their mastery in skills no matter where they were acquired" (Blot, 2014, para. 4; see also Hughes & Dobbins, 2015).

Educational Examples

Digital badge research began around 2010; however, the use of badges is in a relative state of infancy compared to other teaching techniques (Gibson, Ostashewski, Flintoff, Grant, & Knight, 2015). In fact, only one in five institutions of higher education are using digital badges as a means of alternative credentialing (Fong, Janzow, & Peck, 2016).

Research on digital badges has shown mixed results. For example, Reid, Paster, and Abramovich (2015) reported that students had a generally positive view of badges in English courses but the badges did not necessarily enhance motivation. In another study of 100 nursing students, the majority of students reported being motivated by digital badges; yet, the qualitative perceptions of digital badges were mixed and ranged from neutral to positive (Foli, Karagory, & Kirby, 2016). Henry Jenkins (2012), Professor at the University of Southern California noted that badges run the risk of contributing to the "gamification" of education. Jenkins is concerned that badges might become just another system of points. Additional research has shown that some students want to accumulate badges rather than associate learning to them (Resnick, 2012). Abramovich (2013) found that although digital badges improved student interest, they had little effect upon overall competency. Additionally, some scholars noted that such rewards are only effective in the short term and positive effects such as engagement and motivation may decrease over time (Deci, Koestner & Ryan, 2001). Denny (2013) suggested that there is little evidence to support the use of badges to motivate or engage learners.

Badge Credentials in the Workforce and Community

As the open badge movement continues to rise, digital badges may help bridge the gap between the classroom and the workplace. For example, in September of 2014, the National Occupational Competency Testing Institute (NOCTI), the largest provider of industry and partner-based certifications for career and technical education (CTE) programs across the nation, launched the College Credit Recommendation SkillBadge. Students who score at least a 70% on a benchmark approved for college credit by the National College Credit Recommendation Service (NCCRS) earn the SkillBadge (NOCTI, 2015). As employers seek applicants with specific skills and competencies, digital badges may provide an alternative or enhancement to the traditional resume, "By leveraging interest-driven learning and recognizing skills and competencies wherever they are acquired, badges can illuminate unique and personalized pathways to job, career and civic success" (Reconnect Learning, 2014, para. 1).

Badges are also finding their way into community organizations. For example, informal learning spaces like the Dallas Museum of Art (DMA) have incorporated the use of digital badges into a loyalty program to motivate their visitors to become more engaged with the exhibits (See Figure 2). In turn, the badges earned can be redeemed for rewards at the museum; thus, "By creating mutually reinforcing systems and platforms that incentivize engagement, museums can begin to build real datasets of participation to study and learn from" (Stein & Wyman, 2013, para. 52).

Figure 2. DMA Digital Badge (© 2014, DMA. Used with permission.).

According to Schenke (2013), “A systematic study of the motivational impacts of badging has yet to be conducted” (para. 1). Abramovich (2013) suggested that more work is needed to understand the motivation factor of badges across all levels of education and varied learning environments. The purpose of this study was to examine the use of digital badges in higher education across four interdisciplinary areas: education, engineering, human resource development, and nursing.

Conceptual Framework

The conceptual framework guiding the research on using digital badges was based on Vroom’s (1964) expectancy theory. Expectancy theory is woven with underpinnings of behavioral psychology based upon incentives and rewards. Expectancy theory has three main components. 1. *Expectancy*: One believes their effort will result in meeting a goal; 2) *Instrumentality*: Meeting the goal will result in a reward; and 3) *Valence*: The value of the reward is attractive to the individual. This model is based upon human motivational theories and suggests that human motivation is influenced by a particular outcome and the interest a person has in that outcome (Redmond, 2010). For example, in academics, the instructor would present a challenging assignment to the student, then the student will put forth the effort to meet the performance goal if they feel the outcome or reward is worth it (Hancock, 1995).

Methodology

In 2015, as part of a digital badge pilot program, faculty members integrated digital badges into their curriculum. Faculty, from four disciplines (education, engineering, human resource development, and nursing) at a regional university in the Southwestern United States, used the university learning management system (LMS) Blackboard to create digital badges through online achievements. The LMS provided an initial set of badges (see Figure 3) which were set up to automatically deploy to students when key learning milestones were met. However, instructors could create and upload additional badges to the LMS for distribution.

Figure 3. Example of Blackboard Digital Badges (© 2016, Blackboard. Used with permission.).

Furthermore, each badge contained information about the requirements for the credential, the badge distribution date, and the name of the issuer. The students were able to share their awards across social media sites to their peers, friends, and even potential employers.

Research Design

This multi-case study incorporated an exploratory mixed-methods design. A mixed methods study involves the collection and analysis of quantitative and qualitative data in a single study where the data are integrated in the process of the research (Creswell, 1994; Johnson & Onwuegbuzie, 2004). In this mixed-methods study, the quantitative data included demographics, and the qualitative research explored the students' experiences. To accomplish the purpose of the study, answers were sought to the following research questions:

1. To what extent do digital badges motivate students?
2. What are the perceptions of students of digital badges for use in higher education?

Instrumentation

Data was gathered in the form of a pre and post-survey in order to examine the perceptions of students receiving a digital badge in the LMS Blackboard. The pre-survey captured student demographic information such as gender and ethnicity. The survey also asked students the generation they were born in, their school classification, if they played computer games, and whether they had seen a digital badge before. Additionally, using a Likert Scale of 1-5, with 1 being least important and 5 being most important, students rated the importance of making good grades, learning the course content, or being rewarded with tangible items (see Table 1). Of those students that recognized the badging concept, they indicated that locations where they have seen a digital badge included: social media such as Twitter and mobile applications (Swarm, Four Square, Viber, Lose It), an online multi-player game platform (Steam), and in another course.

Table 1. Demographic and Descriptive Data of Participants by Discipline (Cases 1-4) Pre-Survey

	Education	**Engineering**	**HRD**	**Nursing**
Number (N)	35	14	28	13
Ethnicity				
African-American	2	0	13	1
Asian	2	1	1	1
Caucasian	27	12	13	11
Hispanic	4	0	1	0
Other	0	1	0	0
Gender				
Female	35	4	4	12
Male	0	10	24	1
Other	0	0	0	0
Generation (%)				
Silent Generation (1925-1945)0	0	0	0	0
Baby Boomers (1946-1964)	2	0	0	0
Generation X (1965-1981)	5	0	15	7
Millennials/Generation Y (1982-1999)	28	14	13	6
Classification				
Freshmen/Sophomore	0	0	0	0
Junior/Senior	35	14	0	7
Graduate	0	0	28	6
Motivation (Mean)				
Grades	4.74	4.71	*	4.85

Learning Course Content	4.83	4.64	*	4.62
Tangible Items	1.83	1.80	*	1.82
Knowledge of Digital Badge				
Prior Knowledge	1	0	2	3
No Prior Knowledge	34	14	26	10
Computer Gaming				
Played Games	7	6	6	7
Did Not Play Games	28	8	22	6

The post-survey contained four demographic questions, three Likert-Scale questions designed to look at the extent of student motivation after receiving a digital badge, and three open-ended questions to uncover how the badge motivated students and the impact they had in the classroom. Likert scales were all based upon 5-points ranging from (1) "not" to (5) "extremely". The research was approved by the University Institutional Review Board (IRB) prior to data collection.

Analysis of the Data

Data from each case was coded and analyzed independently. The quantitative data from the Likert scales were evaluated using Analysis of Variance (ANOVA) tests to explore differences in motivation among the four disciplines. Analysis of the qualitative data focused on analyzing patterns across the open-ended questionnaire items. According to Stake (2006), there are three main criteria for examining cases: relevance of each case, diversity of each case, and choosing cases that provide the opportunity to learn both "complexity and contexts" (p. 23). In this study, a multiple-case analysis was performed using data across all four cases to look for both similarities and differences. Combined analysis results are discussed below in the cross-case findings corresponding to the two primary research questions in the study.

Data Collection: Cases of the Study

The data for this exploratory study was collected from the following four disciplines: Education, Engineering, Human Resource Development (HRD), and Nursing. The context and post-test results for each case is described next.

Case Study One: Education

An exploratory case study was conducted in an undergraduate course in classroom management within the College of Education and Psychology. Thirty-five female students were told they would receive a digital badge for obtaining at least ten points on each of the seven sections of an online classroom management portfolio. Each section was comprised of a written paper and three supporting artifacts.

Post-Survey

Twenty one (60%) of the students completed the follow-up post-survey. All of the students reported receiving at least one digital badge in their course. Using a Likert Scale, students were asked whether receiving a digital badge in their course motivated them. Students (n=19) reported an average of 3.16 (SD=1.21). When asked why they were motivated to receive a badge, students responded with comments such as: “[It] visually stimulated my motivation to do well”, and “I liked seeing it when I would log in or as a notification on my phone”.

When students were asked how motivated they were to work and receive other badges, they reported an average of 3.22 (SD=1.17). One student noted, “It wasn't like I was just being graded and moving along, the instructor actually took the time to go in and give that badge to me”. And, when students were asked to what extent did getting a badge motivate you to work harder, the average was a 3.17 (SD=1.20). One student remarked, “It made me feel like my work was really good and that all the hard work was being appreciated”. Another student noted, “They showed me that I was doing a good job and made me want to continue to do well”. When students were asked "Do you believe that digital badge systems can have a positive impact on students?" 17 reported yes and 3 reported no. For the 15% of students who noted badges did not motivate them, the emphasis was on the receiving a good grade in the course as illustrated in the following two excerpts: "The college degree is the motivator" and "The badges did not necessarily motivate me to work harder; they were just like the little cherry on top for receiving a good grade."

Case Study Two: Engineering

Twenty-nine undergraduate students (junior and senior level) enrolled in a required Engineering Hydrology and Hydraulic Structures course participated in this study. The course, which was offered through the Department of Civil & Environmental Engineering, consisted of two 55 minute lectures and one 2 hour, 45 minute laboratory per week.

Students received digital badges for attending laboratory classes on different topics. Each instructor made badge (see Figure 4) was worth 20 points (or one percent) towards the student's final grade; there were a total of 10 badges that could be earned. And, full credit was awarded for attending the lab, participating and following all of the safety regulations of the lab. The badges were awarded through Blackboard and placed in a folder so the students could see how many badges they earned and the attendance points were uploaded to the online gradebook.

Figure 4. Example of an Instructor Made Badge

Post-Survey

Eleven students (10 male, 1 female) completed the post-survey. All of the students reported having received at least one digital badge in this course. Using a Likert Scale of 1-5, the students were asked "how motivated were you by receiving a digital badge in your course" and the students reported an average of 3.31 (SD=1.13). The same scores were reported when students were asked to “what extent were you motivated to work and receive other badges”. When asked “did getting a badge motivate you to work harder”, the students reported an average of 3.00 (SD=1.07) (see Table 1). In fact, one student remarked that digital badges, “showed me that I could, with enough effort that I could receive an A”. When students were asked “Do you believe that digital badge systems can have a positive impact on students?” 6 reported yes and 5 reported no. One student stated, “It doesn’t help you in a class”.

In general the engineering students were neutral as to whether or not digital badges motivated students. Based on the open-ended comments, the students felt that if the badges helped your grade/GPA, badges would be useful. In this course, badges were not given for additional work in the class (e.g. extra credit) but a visual reminder of the importance of laboratory attendance. The badges may have had more meaning to students if the badges were directly tied to the grade.

Case Study Three: Human Resource and Development

An online graduate business course focused on workforce and talent development. During the semester course, the graduate student participants utilized multiple learning techniques to explore the use of digital badges as an alternative credentialing method in organizations. First, the graduate students who completed the study (N=20) were required to read four resources about digital badges and alternative credentials published online (Educause, 2012, 2013; Fleming, 2015; Parker, 2015).

In addition, the instructor offered the graduate students an opportunity to earn digital badges in their course management system (Blackboard) conditional on their earning a grade of at least 90% on targeted assignments by a specific due date. Further, the students were asked to offer a professional context for using a digital badge and describe that information in a semester research paper on digital badges as an alternative credential technique in the workplace. The students were directed to create a sample digital badge for their developed context and then embed the digital badge within their research paper as a figure, describing the context as well as including at least six sources with a minimum of at least three sources being peer-reviewed journal articles. One example of a student paper on digital badges explored the notion that a digital badge might be utilized to reward employees who successfully complete a number of specified leadership courses online and within training sponsored by their organization. By doing so, the student remarked that an organization benefits when it "develops future leaders within the organization… [that] adds value to their current talent pool". The badge is a visual recognition for those employees who pursue leadership development which in turn adds value to the organization where they work (Figure 4).

Figure 4. Example of a Digital Badge

Post-Survey

For the final task of the course, the graduate students were also administered a post-survey. Results from the survey reflected that students were moderately motivated to receive a digital badge in the course (on a scale of 1-5, the average score was 3.85; SD=.99). When asked why they were motivated to receive a badge, descriptive comments from students included responses such as: "It gave me an incentive to push forward", "After receiving my first badge, I realized that by getting my assignments in on time, I was being rewarded...like a sticker for well done"; and, "It felt good, I laughed though it may look like something for children, but I realized it had effect on me to work harder. It is like a kind of kudos, or a pat on the back". Thus, the majority of the graduate students reported that they felt positive toward earning a digital badge themselves in a graduate course. In addition, when asked if they felt that digital badge systems can have a positive impact on students, 95% (18 of 19 students) answered yes (See Figure 5).

Figure 5. Data based on a Post-Experience Survey on Digital Badges

When asked why (or why not) digital badge systems can have a positive impact on students, participant remarks included, “Helps show achievement and skills earned”, “Could definitely encourage”, and “a form of recognition and motivation”. No negative comments were given by the participants in response to this question. The researcher concluded from this pilot study in a graduate business course that digital badge systems have the potential to motivate graduate students in online courses as well as the potential to motivate employees in organizations.

Case Study Four: Nursing

Nursing students enrolled in three required nursing courses (Community Health Nursing, Health Assessment, Education Roles and Practicum) participated in this study. Community Nursing, a hybrid undergraduate course, met online and face to face one day a week on alternating weeks. Badges were awarded for a grade of 90 or higher on two quizzes administered during the course. Students received a badge if they met the requirements, but they were not informed they would receive the badges before completing the two assessments.

Health Assessment was a web-based lecture course with a weekly three-hour laboratory for undergraduate students. Badges were awarded for completing course forms and for completing three online modules by the due date. The Education Roles and Practicum was an online, graduate course where digital badges were awarded for completing preceptor information in a timely manner. Badges were awarded through Blackboard but students were not informed they would receive badges ahead of time.

Post-Survey

Twelve female students completed the post-survey (two graduate students and 10 undergraduate students). When asked if the student received a digital badge in a course, three said yes and nine said no, demonstrating a lack of awareness of the badge. On a scale of 1-5 (with 1 being not motivated to five being extremely motivated), three students answered with a mean motivation level of 4. The same result occurred for the question “To what extent were you motivated to work and receive other badges?”

Based on the low and incomplete response rate, nursing students do not appear to be motivated by a badge alone. This may be for a variety of reasons, but it may relate to the perceived limited use of badges in Nursing work settings. If badges were seen as something that students could utilize for career developing such as adding to their resume, an ePortfolio, and help them get a job, perhaps they would be more motivated to receive a badge.

Cross-Case Findings

To answer the two research questions in this study, we looked across the data and analyses from the four cases for shared patterns and characteristics. Across the cases, 94% of the students taking the pre-survey reported having never seen a digital badge before.

Descriptive statistics were used to answer research question one: *To what extent do digital badges motivate students?* Table 2 shows the difference in mean and standard deviation for each subject area per motivation area. As noted above, measures in motivation are value ranges from 1 to 5 on a Likert Scale (with one representing low to five representing high).

Table 2. Differences in motivation by group

Motivation	**Education**	**Engineering**	**HRD**	**Nursing**
	(*N* = 21)	**(*N* = 29)**	**(*N* = 28)**	**(N=12)**

6. Did you receive a digital in the course?	3.16 (1.21)	3.13 (1.13)	3.85 (.99)	4 (1)
7. To what extent were you motivated to work and receive other badges?	3.22 (1.17)	3.13 (1.13)	3.50 (1.16)	4 (1)
8. To what extent did getting a badge motivate you to work harder	3.17 (1.20)	3.00 (1.07)	3.62 (1.26)	4 (1)

An Analysis of Variance (ANOVA) was conducted on paired and grouped data sets. The confidence interval was set to 95% (alpha = 0.05). Each of the four disciplines (engineering, education, nursing, and HRD) were compared to each other. There was no statistical difference found between the student perceptions in any of the four disciplines for questions 6, 7 and 8. However, there were statistically different student perceptions about digital badges for Question 10: *Do you believe that digital badge systems can have a positive impact on students?* Nursing and Engineering majors seemed less convinced that digital badges could have a positive impact on student learning. To further test this hypothesis, the data was grouped into applied majors (Engineering and Nursing) and non-applied majors (Education and HRD) and a statistical difference was found for the aforementioned question.

Data was grouped into male and female students for all four disciplines. As previously found, there was no difference for Questions 6, 7 and 8, but there was a difference in Q10 (*Do you believe that digital badge systems can have a positive impact on students?*). Male students were more likely to comment that they did not believe digital badges could have a positive impact on students.

Table 2. Summary of ANOVA results for Q10

	Education	HRD	Nursing	Non-applied	Male
Engineering	0.007	0.007	0.746		

Education		1.0	0.032		
HRD			0.032		
Applied				0.00036	
Female					0.011

RQ2. What were the perceptions of students of digital badges for use in higher education?

Analysis of the qualitative data focused on analyzing patterns across the three open-ended questionnaire items. In regards to Research Question 2: *What were the perceptions of students of digital badges for use in higher education?*, five broad themes emerged: Acknowledgment, Internal Motivation, External Motivation, Competition, and Accountability (See Table 3).

Across all four disciplines, students noted that badges were an acknowledgement of the work they had done. Education, Engineering, and HRD students noted that digital badges were externally motivating. However, several of the students indicated that the grade was more motivating than the badge. For example, an Education student remarked, “They didn’t really motivate me honestly; I just wanted a good grade”. Neither Engineering nor Nursing students conveyed that badges were internally motivating to them. Furthermore, both Education and HRD students reported that badges were somewhat of a competition amongst the students. Finally, it should be noted that both Engineering and HRD students stated that badges helped to keep them accountable.

Table 3. Perceptions of Digital Badges

Themes	**Case One Education**	**Case Two Engineering**	**Case Three Human Resource Development**	**Case Four Nursing**
Acknowledgment	It is recognition of our work.	It's like a reward for doing work.	Badges are a form of recognition and motivation	It is a nice reward for a completed job.
Internal Motivation	It makes the student happy that		It felt good, I laughed though it may look like	

	they did a good job. It just made me feel good so I think it does the same to others also.		something for children, I realized it helped me to work harder. Gave me incentive to push forward	
External Motivation	The badges are along the lines of check marks or grades and most students want to please the teacher. Even though it's a small thing, like getting a sticker in elementary school, it does give you a little extra pride in your work. Some students may need the badges as an incentive to work hard.	If you keep receiving good grade it's like an incentive to keep up the hard work and get an A in the class If each badge counts for points towards a test or a final grade it will inspire the students to do whatever it takes to get a badge.	I wanted to make sure I edited my papers better and researched more. It was an award and I wanted one. After receiving my first badge, I realized by getting my assignments in on time, I was being rewarded. I was motivated because I was rewarded for my service learning assignment.	
Competition	It is almost like a contest to see who can get the most badges.		It made me feel like I stood out more by having digital badges	
Accountability		Helps keep track of grades. Because it keeps you up to date on how you're doing.	Helps to show achievement and skills learned.	

Discussion and Conclusion

In this research study, we explored the perceptions of 90 students across four disciplines. There were differences between the student sets with the applied major students seeming to be less interested in badges as a motivational tool than the non-applied majors. For example, when students were asked "*Do you believe that digital badge systems can have a positive impact on students*?" 90% of Education students and 90% of HRD students responded yes, while only 55% of Engineering students and 62% of Nursing students responded yes. According to Fanfarelli and McDaniel (2015), individual differences among students can affect the way that receiving badges is perceived. Our results suggest that groups of students show differences in their opinions toward the use of badges in education. For example, students with applied majors tended to be less compelled by the utility of badges while those in non-applied majors viewed them more favorably. These results show that, while digital badges may be worthwhile in some disciplines, they may not be worthwhile in others. We also found evidence for differences in the ways that students responded to badges as a function of gender. Taken together, there is evidence that previous studies finding mixed evidence on the utility of the badges may be due in part to unexplored group differences within the sample. Students in all four disciplines favored the use of digital badges whether or not the students thought that they were motivational. Research (UPCEA, 2017) also indicated that millennials valued badging; however, regardless of discipline, all three generations (Baby Boomer, Generation X, Generation Y) of participants surveyed reported that tangible items like digital badges were less motivating to them than receiving good grades and learning the course content.

Beyond differences in student compositions, it may be possible that these results were due to the way in which the tested courses implemented badges. Across all disciplines, faculty members used or created digital badges for use within the LMS; however, only HRD students created their own badges to display as part of their course assignment, perhaps creating more personalized learning and leading to the higher percentage of impact. Again, at least one issue with the current evaluation of the efficacy of badges in education is that there remains no standardized method of implementing them into courses. This was also seen in research by Abramovich (2013), Deci, Koestner & Ryan (2001) and Denny (2013) in which badges may have had a small influence in student interest but overall were not tied to competency of the material or motivation. A study conducted by Fajiculay, Parikh, Wright, and Sheehan (2017) with pharmacy students supports these findings as students perceived badges to be helpful for learning course material. While the badges may not motivate the students to complete the assignments, some students did seem to appreciate the extra attention they received for completing the assignment; the assumption from the comments is that students would have completed the assignment for the grade whether or not there was an opportunity to receive a digital badge.

The data revealed the need for the instructors to more clearly define and communicate the requirements for digital badges to the students. Motivation may have depended upon the quality control of these awards and as faculty learn to implement them into their courses, there may be an increase in student motivation, both externally and internally. To further assess the value of digital badges it would be necessary to expand digital badges into more classes within a single discipline or over multiple disciplines. And, increasing the exposure of students to digital badges in a single discipline would further define student perceptions. Additional research is needed to determine if digital badges motivate students more when associated with a grade or work outside classroom than as a reward for an assignment for which they already earned credit. Furthermore, the badging done in this study was not sharable across the LMS as part of an open badging system.

Students want access to a digital campus (DJS Research, 2016). Digital badges are a component of e-credentials, like e-portfolios and expanded transcripts that display students' projects; and are part of a foreseeable academic movement toward enhanced digital processes (Hickey, 2017). Indeed, the future of digital badges, or alternative credentials, reaches further than the higher education classroom and some universities are promoting competency badges that could provide proof of skills needed for various jobs (Bull, 2015). If we truly advocate life-long learning, additional research should be done on how badging can move beyond participation and assignment completion to true credentials of mastery learning.

Limitations

There were several limitations to this study. One limitation of this study is its generalizability due to the fact that the study was based on just 90 students from one four-year institution. In addition, the pre-survey cannot be associated with all of the specific students in the post-survey as the surveys were anonymous. Nor were students required to answer any of the questions which led to some variability in responses. An additional limitation in this study is that the faculty were new to badge creation and implementation. Some of the badges were awarded for meeting some criterion (performance-based), while other badges were awarded for effort (Schenke, 2013). Future research needs to be conducted to determine differences amongst graduate and undergraduate students by focusing on similar disciplines. There were not enough graduate or undergraduate students in any one discipline to further analyze this difference. The different types of badges may have affected student motivation. And, as noted previously, some students were not aware that they could earn badges.

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